Universidad Nacional Abierta y a Distancia

Vicerrectoría Académica y de Investigación

Learning Guide for the Development of the Practical Component of the Course

1. General Information about the Practical Component

Table 1. *General Information.*

Aspect	Description
1. Methodological strategy	Task-Based Learning
2. Course typology	Methodological
3. Evaluation moment	**Intermediate**
4. Activity score	**150**
5. Number of activities of the practical component registered in this guide	**1**
6. Student independent work hours	**56**
7. Hours of teaching support	**24**
8. Type of formative practice	**Simulated**

2. With this/these practical component activity/activities, it is expected that students achieve and demonstrate the following learning outcome(s):

Apply supervised and unsupervised models in machine learning for decision-making.

General description of the activity/activities of the practical component.

Table 2. *Activity 1 Information.*

Aspect	Description
1. Scenarios of the practical component	**Simulated with ICT Support**
2. Type of activity	Independent
3. Activity number	1
4. The activity starts on:	Wednesday, February 18, 2026
5. The activity ends on:	Tuesday, March 17, 2026

The following materials and resources are required for the development of this activity:

- Desktop or laptop computer. Internet connection.
- Python software installed on the computer or Google Colab account.
- Deepti, C., & Roopal, K. (2023). Introduction to Machine Learning with Python. Bentham Science Publishers. (Sections Linear Regression and Logistic Regression, Decision Trees y Supervised Learning).
- Colliot, O. (Ed.). (2023). Machine learning for brain disorders (Neuromethods, Vol. 197). Humana Press. (pp 25-60).
- Appendix 1 – Google_Colab.pdf
- Appendix 2 – Template_Phase2.ipynb

The activity consists of:

For the development of this guide, each student must select an item A, B, C, D or E and develop exercises 1, 2 and 3 according to the selection. In addition, each student must express their choice in the Phase 2 forum through the following table

Table 3. *Exercise item selection*

Student Name	Ítem
	A
	B
	C
	D
	E

Exercise 4 consists of selecting a literal developed by one of their classmates and giving feedback. Next, you must present in the forum the suggestions and/or adjustments to be made identified in the development presented by your partner.

You must submit a single document with the 4 exercises developed in the format **Appendix 2 – Template_Phase2.**

<u>Exercise 1: Contextualization Dataset</u>
According to the selection in **Table 3**, each student must download the datasets described in the corresponding item and in a markdown cell write in their own words the respective information:

- Context datasets.
- Sizes.
- Variable objectives.
- Associated variables.

A: Import the live-disorders dataset (ID:8) for Regression and credit-g (ID:31) for Classification.
B: Import the cholesterol dataset (ID:204) for Regression and blood-transfusion-service-center (ID:1464) for Classification.
C: Import the cloud dataset (ID:210) for Regression and phoneme (ID:1489) for Classification.
D: Import the bodyfat dataset (ID:560) for Regression and Diabetes (ID:37) for Classification.
E: Import the cleveland dataset (ID:194) for Regression and steel-plates-fault (ID:1504) for Classification.

Exercise 2: Supervised Learning.

Using the train_test_split command, divide the dataset into 75% for training and 25% for testing, and apply the following models for regression dataset:

1. Linear regression.
2. Lasso.
3. Ridge.

Make a table comparing the results of MSE, MAE, RMSE, and R2 metrics. Explain in a markdown cell what is presented in the table.

Apply the following models for classification dataset:

1. Logistic regression.
2. Decision tree.
3. Perceptron.

Make a table comparing the results of Accuracy, Precision, Recall, and F1-score metrics. Explain in a markdown cell what is presented in the table.

Exercise 3: Unsupervised Learning.

According to the two (2) datasets corresponding to the item chosen in the **Table *3***, apply instance clustering algorithms and establish profiles for each cluster.

Note that you must perform all activities in Python and to the two (2) datasets.

Apply the k-means algorithm by selecting two numerical variables that you consider most relevant and solve the following literals based on this:

A. Perform data cleansing by removing NaN, outliers and standardizing data, etc.

B. How many clusters would it be appropriate to select to represent the data?

C. What criteria do you use to determine this k-number?

D. Plot a scatterplot on the two chosen variables, identifying the chosen clusters

Reapply the k-means algorithm, increasing the number of numerical variables and solve the following literals based on it:

E. Does increase the number of variables improve cluster clustering? To do this, check the metrics in the elbow method and Silhouette Score.

With the best algorithm obtained (with 2 variables or with more variables) determine in detail, in your words, the characteristic profile of each cluster in terms of the variables provided.

Ejercicio 4: Feedback

Select a literal developed by one of their classmates and give feedback. Attach a screenshot of the feedback given to one of your peers on the Phase 2 forum.

Evidence of independent work:

The individual work evidences to submit are:

- The contribution of the 3 exercises published in the forum must be in PDF or JPG format. In the Evaluation environment, the .ipynb file will be attached in the format found in the folder Guide for the development of the practical component and evaluation rubric - Unit 1 - Phase 2 - Practical component - Simulated practices (**Annex 1 - Phase 2 Template**), to apply supervised machine learning models.
- Delivery of a file in .ipynb format of the exercises developed and named G#_(name_last name)_Phase2, for example:

G15_Rafael_Gaitan_Phase2.ipynb

Evidence of group work:
No group work evidences are required for this activity.

1. **General guidelines for the elaboration of evidences**

For evidence developed Choose an option. Please consider the following guidelines:

All group members must participate with their contributions in the development of the activity.

Each group should designate one member to be responsible for submitting the requested product in the environment specified by the teacher.

Before submitting the requested product, the group should check that it complies with all the requirements indicated in this activity guide.

Only group members who have actively contributed during the allocated time for the activity should be included as authors of the requested product.

Please note that all individual or group written products must comply with "spelling and mechanics" standards and the submission conditions defined in this activity guide.

Regarding the use of references, consider that the product of this activity must comply with the Choose an option standards.

In any case, comply with the referencing standards and avoid academic plagiarism. To do so, you can support yourself by reviewing your written products using the Turnitin tool available in the virtual campus.

Consider that in Agreement 029 of December 13, 2013, Article 99, the following actions are considered as offenses against academic order, among others: item e) "Plagiarism, that is, presenting as one's own the entirety or part of a work, paper, document, or invention created by another person. It also includes the use of false citations or references, or proposing citations where there is no match between the citation and the reference," and item f) "Reproducing or copying, for profit, educational materials or results of research products that have intellectual property rights reserved for the University."

The academic sanctions that the student will face are as follows:

a) In cases of proven academic fraud in the respective academic work or evaluation, the grade imposed will be zero points without prejudice to the corresponding disciplinary sanction.

b) In cases related to proven plagiarism in academic work of any nature, the grade imposed will be zero points, without prejudice to the corresponding disciplinary sanction.